

Hi [AD]

I'd like to share my thoughts on how the team should proceed in testing the hypothesis that churn is driven by customers' price sensitivities. To test the hypothesis, we would need to model the churn probabilities of customer and obtain the effect of prices on churn rates. There are three datasets we need to build the model. This is the following data we needed:

1. Customer Data: This dataset should include relevant information about customers, such as their joining date, historical electricity usage, and other pertinent details.
2. Churn Data: We need a dataset that indicates whether customers have churned or not, allowing us to determine the churn status for each customer.
3. Pricing Data: This dataset should encompass historical pricing information, including the prices charged to each customer over time.

After we have the data, we can perform Exploratory Data Analysis (EDA) to gain insights and confirm whether churn is influenced by customer price sensitivity. Since this is a classification problem, we will build binary classification models such as Logistic Regression, Random Forest, and Gradient Boosted Machines to name a few. We will select the most suitable model based on considerations such as model complexity, interpretability, and accuracy.

With the selected model, we can determine both the direction and extent of the influence of prices on churn rates. Additionally, we can assess the significance of prices in relation to other factors. Moreover, the model will enable us to estimate the business implications of the client's proposed discounting strategy.

Best regards,
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